Fault-Tolerant Motion Planning for Rocket-Propelled Spacecraft: Actuator-Fault Impacts in Capability Space and Along an Apollo Lunar Module Descent

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Abstract

An actuator fault changes both what a spacecraft is able to produce at an instant and whether it can still complete its manoeuvre. The first question belongs to control-allocation theory, the second to trajectory optimisation, and the two are seldom asked of the same vehicle and the same faults. This work measures both. Faults are first expressed as perturbations of the attainable force and moment sets of a rocket-propelled spacecraft, which classifies each fault by whether it contracts, expands or leaves unchanged the set of producible wrenches, or removes the zero wrench from it. The classification is applied to the Apollo Lunar Module and then to 18 spacecraft across five mission classes. Survivability is then measured independently, by direct transcription of the six-degree-of-freedom landing problem and solution as a nonlinear program: 800 solves over 16 faults and 5 initial-condition regimes, 576 over engine placement, and 180 faults injected at points along a single Apollo-anchored descent. Every fault examined leaves the Lunar Module's hover wrench attainable, yet four of them have onset points from which no landing trajectory exists. The least survivable fault, a one-interval transport delay, leaves the attainable sets numerically identical to the healthy vehicle. Attainable-set analysis is therefore necessary but not sufficient for fault-tolerant design: it describes what a damaged vehicle can produce, and not whether it can produce it in time.

Keywords: fault-tolerant control, control allocation, attainable moment set, trajectory optimisation, powered descent, actuator redundancy

Nomenclature

Symbol	Definition
$\mathbf{F}, \mathbf{M}$	net force and net moment on the vehicle, body frame (N, N m)
$\mathbf{w}$	wrench, the stacked pair $(\mathbf{F}, \mathbf{M}) \in \mathbb{R}^6$
$m, \mathbf{J}$	vehicle mass and inertia tensor (kg, kg m^2)
$\mathbf{v}, \omega$	body-frame velocity and angular rate (m/s, rad/s)
$\mathbf{d}_i, \mathbf{r}_i$	unit thrust direction and position of actuator i
$\mathbf{B}$	allocation matrix mapping actuator commands to wrench
T, T_{min}, T_{max}	engine thrust and its throttle limits (N)
η	thrust efficiency; delivered thrust is ηT
δ, δ_{max}	gimbal deflection and its limit (deg)
ω_n, ζ	natural frequency and damping ratio of the gimbal servo
τ_T	time constant of the engine thrust response (s)
y_{eng}, dz_{eng}	engine half-spacing and axial offset from the centre of mass (m)
t_f, τ_d	fault onset time and planner reaction delay (s)
g	lunar gravitational acceleration, 1.625 m/s^2

Term	Definition
6-DoF	six degrees of freedom: three translational, three rotational

Term	Definition
RCS	reaction control system; the tier of small, fixed, one-sided thrusters
TVC	thrust vector control; steering of the main engine by gimbal
DPS	descent propulsion system, the Apollo Lunar Module main engine
FTC	fault-tolerant control
OCP, NLP	optimal control problem; nonlinear program
One-sided actuator	an actuator that can push in one direction only, $f_i \geq 0$
AFS	attainable force set: all net forces the actuators can produce
AMS	attainable moment set: all net moments the actuators can produce
Capability space	the space in which the AFS and AMS live; the space of producible wrenches
Conditional AMS	moments attainable while the vertical force is held at the hover value
Hover wrench	the wrench that holds the vehicle in steady hover: weight up, zero moment
Support function	$h(\mathbf{u}) = \max\{\mathbf{u} \cdot \mathbf{x} : \mathbf{x} \in X\}$, the reach of set X in direction $\mathbf{u}$
Mean support radius	the average of $h(\mathbf{u})$ over directions; a size measure valid for flat sets
Glide cone	a minimum elevation angle of the vehicle above the horizon, seen from the landing site
Landing gate	the fixed set of touchdown limits a trajectory must satisfy to count as a landing
Gate margin	worst gate criterion divided by its limit; ≤ 1 is a landing
Transport delay	a fault that shifts the whole control sequence by one discretisation interval

I. Introduction

A rocket-propelled spacecraft controls its position and attitude with a small number of actuators that push in one direction only. A main engine produces thrust along its nozzle axis and can be gimballed through a few degrees; a reaction-control thruster either fires or does not. When one of these actuators degrades, two distinct questions arise, and they are conventionally answered by different methods. The first is what the vehicle can still produce: the set of net forces and moments available from the surviving actuators, studied in the control-allocation literature as the attainable force and moment sets [15, 16, 18]. The second is whether the vehicle can still complete its manoeuvre: whether a dynamically feasible, constraint-satisfying trajectory to the goal still exists, which a trajectory optimiser decides [4, 5, 12]. The first is a statement about an instant, the second about a sequence.

This thesis measures both on the same vehicles and the same faults. Three contributions follow.

1. A characterisation of actuator faults in capability space. Each fault is expressed as a perturbation of the attainable force and moment sets and classified by its effect on them.
2. A fleet survey applying the same measures to 18 spacecraft in five mission classes, which tests whether the single-vehicle classification is a property of the faults or of the vehicle.
3. A quantitative evaluation of fault-tolerant motion planning for an Apollo-class Lunar Module, comprising 1,556 six-degree-of-freedom optimal-control solves across four campaigns.

The vehicle of record is the Apollo Lunar Module in its powered-descent configuration. It was chosen because its mass, inertia, thrust, gimbal throw and reaction-control layout are documented in the public record [24, 26], and because a lunar landing has an unambiguous success criterion. Fault detection, isolation and identification are outside the scope: every result assumes the fault becomes known to the planner at the instant it occurs, except in the one campaign that varies the reaction delay deliberately. Consequently, every survivability figure reported here is an upper bound on what a system with a realistic diagnosis latency would achieve.

Section II reviews the two literatures this work sits between. Sections III–VI give the method: the vehicle and actuator models, the fault model, the simulation and motion-planning environment, and the design of the evaluation campaigns. Section VII reports the results, Section VIII interprets them, and Section IX concludes. Every figure was produced by a script in the accompanying repository; Appendix A maps each result to its script.

II. Background and related work

A. Six-degree-of-freedom spacecraft dynamics

The rigid-body motion of a spacecraft is described by Newton's law for the translation of the centre of mass and Euler's equations for rotation about it. Standard treatments are Wie [1], Hughes [2] and Markley and Crassidis [3]. In the body frame,

$$m(\dot{\mathbf{v}} + \omega \times \mathbf{v}) = \mathbf{F}, \qquad \mathbf{J}\dot{\omega} + \omega \times \mathbf{J}\omega = \mathbf{M}.$$

Two properties of this system matter throughout. The vehicle is underactuated in the wrench: its actuators cannot produce arbitrary $(\mathbf{F}, \mathbf{M})$ pairs, because thrusters are one-sided and engines are confined to a narrow cone about the nozzle axis. And force and moment are cross-coupled: an engine offset from the centre of mass cannot produce force without also producing moment, so a thrust fault is never only a thrust fault.

B. Motion planning by constrained optimisation

Trajectory generation for such systems is posed as an optimal control problem and solved numerically. Betts [4, 5] distinguishes indirect methods, which solve the optimality conditions, from direct methods, which discretise the trajectory and pass the resulting finite problem to a nonlinear-programming solver. Direct transcription [6, 7] dominates practice because path constraints enter as ordinary algebraic inequalities.

Powered descent has a substantial convex-optimisation literature. Açıkmeşe and Ploen [8] cast Mars powered descent as a convex program; Blackmore et al. [9] extended it to minimum-landing-error guidance; and Açıkmeşe et al. [10] proved lossless convexification, under which the non-convex lower thrust bound can be relaxed without changing the optimum. Full 6-DoF landing with attitude states is not convex, and is treated either by successive convexification [11] or by direct transcription to a general nonlinear program. Malyuta et al. [12] survey the field.

The present work uses direct transcription with a general-purpose interior-point solver rather than a convexified formulation, because several of the faults studied here alter the structure convexification relies on: a valve stuck open removes the feasibility of the lower thrust bound, and a transport delay shifts the control grid. The problem is built with CasADi [14] and solved with IPOPT [13].

C. Control allocation and attainable sets

Where the planner demands a wrench, the allocator must produce it from actuators with limits. Durham [15, 16] introduced the attainable moment set, the image of the admissible control set under the allocation map, as the object that decides whether a demanded moment is achievable. Bodson [17] compared allocation algorithms and Johansen and Fossen [18] surveyed the field. For one-sided actuators with bounded magnitude the attainable set is a convex polytope, and testing whether a required wrench lies inside it is a linear program; Sections III and VII use exactly that construction.

Whether a set of one-sided actuators retains authority about every axis is settled by Farkas' lemma [21]: authority is lost precisely when a direction exists that no surviving actuator can oppose. This gives an exact count of which actuator failures cost a control degree of freedom, in place of a worst-case assumption.

D. Fault-tolerant control

Zhang and Jiang [19] classify fault-tolerant control into passive schemes, which are robust to an anticipated fault set, and active schemes, which detect a fault and reconfigure. Blanke et al. [20] give the standard treatment of diagnosis and reconfiguration. The faults modelled here are drawn from the propulsion literature — throat erosion, injector blockage, feed-coupled combustion instability and valve failures [33, 34] — and are expressed in the three dynamical forms fault-tolerant control uses: multiplicative changes to actuator gain, additive disturbances, and structural changes to the input set.

E. Position of this work

The allocation literature measures a fault by its effect on the attainable set. The trajectory-optimisation literature measures it by whether a plan still exists. Both are standard, and they are rarely applied to the same fault set on the same vehicle. Section VIII shows that on this vehicle they disagree.

III. Vehicle and actuator models

A. The fleet and its data

18 spacecraft with usable actuator geometry are modelled, drawn from five mission classes: launch-vehicle boosters, low-Earth-orbit satellites, geostationary satellites, crewed vehicles and deep-space probes. Each vehicle is described

by mass, principal inertia, a list of reaction-control thrusters (position, unit fire direction, rated thrust) and a list of main engines (pivot position, nozzle axis, rated thrust, gimbal cone half-angle).

The underlying parameter workbook supplies thrust magnitudes, actuator counts and masses from published sources. It contains no thruster positions, no thrust directions, no gimbal limits and no inertia tensors; all four are supplied by the geometry model, and each entry carries a provenance flag. The split is 87 sourced, 82 estimated and 1 missing across 175 entries. Only the Apollo Lunar Module layout is a validated model rather than a reconstruction, which is why the detailed fault campaigns are run on it and not on the fleet.

B. Attainable force and moment sets

For a vehicle with one-sided thrusters and gimballed engines, the attainable force set (AFS) and attainable moment set (AMS) are computed by a support-function sweep. For each of a large number of unit directions $\mathbf{u}$, the admissible actuator combination that maximises $\mathbf{u} \cdot \mathbf{x}$ is found in closed form, and the convex hull of those support points is the set [15, 16, 22].

Three size measures are used. Volume is the natural one but vanishes for degenerate sets, and a vehicle with a single centreline gimballed engine has no roll authority at all, so its moment set is planar. The mean support radius, the average of the support function over directions, is therefore used wherever vehicles of different architecture are compared. Containment of the origin is reported separately, because it is a topological property no size measure detects: if the origin leaves the set, the zero wrench is no longer attainable.

The unconstrained AMS overstates what a descending vehicle has available, because it is free to spend the whole actuator set on one moment and let the vehicle fall. The conditional AMS is therefore also computed: the set of moments attainable while the vertical force is held at the value that supports the vehicle's weight, obtained by solving one linear program per direction. Whether the hover wrench itself remains attainable is tested the same way.

C. Redundancy as a geometric property

A count of actuators is a poor measure of redundancy when actuators are one-sided, because the survivors may all push the same way. The exact criterion is Farkas' lemma [21]: a set of thrusters retains authority about every axis unless some direction exists that none of the survivors can oppose. Enumerating subsets and testing this condition identifies the minimal fatal loss sets. The same enumeration over all 2^N subsets of N actuators gives the fraction of the loss space that costs a degree of freedom. That fraction is a property of geometry alone: it assumes no per-unit failure probability, and it is reported as such rather than as a reliability.

D. The Apollo Lunar Module model and its engine spacing

The Lunar Module has one descent engine. To study engine-out faults at all, the model splits it into two half-thrust engines at $y = \pm y_{eng}$. This is the one invented parameter in the vehicle, and its value is bounded. A single surviving gimbal can trim the moment left by a dead engine only while

$$y_{eng} \;\leq\; dz_{eng} \tan \delta_{max} \;=\; 0.263 \text{ m},$$

with $dz_{eng} = 2.50$ m and $\delta_{max} = 6°$ taken from the vehicle [26]. Below this threshold the split adds no roll authority the real vehicle lacked; above it, the model would fly a vehicle with authority the Lunar Module never had. The spacing is fixed at $y_{eng} = 0.25$ m throughout, except in the campaign that varies it deliberately.

IV. Fault model

Faults are expressed as changes to the actuator model, in the three forms used in the fault-tolerant control literature [19, 20].

Multiplicative faults change actuator gain. Thrust efficiency $\eta \in (0, 1]$ with delivered force ηT represents turbopump degradation, injector blockage and throat erosion [33]; $\eta > 1$ represents a pressurant-regulator runaway. A slow thrust response is a change to the lag time constant τ_T, and a sluggish or underdamped gimbal a change to (ω_n, ζ).

Additive faults introduce a term independent of the command: a fixed nozzle misalignment from asymmetric erosion, or a feed-coupled thrust oscillation (chugging) at fixed amplitude and frequency [34].

Structural faults change the input set itself. An engine fails off; a valve sticks open so the engine cannot be throttled below a floor $T_{min} > 0$; a gimbal seizes; a transport delay shifts the whole control sequence by one discretisation interval.

The twelve faults evaluated on the Lunar Module are listed in Appendix B with their class and temporal character. The fleet study uses a nine-fault subset applicable across architectures.

V. Simulation and motion-planning environment

A. Plant

The plant is a 6-DoF rigid body above a flat lunar surface with constant gravity $g = 1.625$ m/s^2, augmented with actuator states. The rigid body carries twelve states — position and velocity in the local frame, Euler angles and body rates — and each engine carries five more: thrust, and pitch and yaw gimbal deflection with their rates. With two engines the model has 22 states.

Engine thrust responds as a first-order lag. Each gimbal axis is a second-order servo, $\ddot{\delta} = \omega_n^2(\delta_c - \delta) - 2\zeta\omega_n\dot{\delta}$, so that a degraded actuator is a change of parameters rather than a change of equation. The 16 reaction-control thrusters are one-sided, $0 \le f_i \le F$, and enter through a fixed 6×16 allocation matrix $\mathbf{B} = [\mathbf{d}_i;\ \mathbf{r}_i \times \mathbf{d}_i]$.

Vehicle parameters are the documented Apollo values [26]: 7,711 kg landing mass, 45,040 N of descent-engine thrust, $\pm 6^\circ$ of gimbal throw, and 16 thrusters of 445 N in four quads. Mass is held constant; propellant depletion is not modelled.

B. Transcribed optimal control problem

The landing problem is transcribed directly. States and controls at every node are decision variables; the dynamics are imposed as equality constraints between consecutive nodes by fixed-step Runge–Kutta integration; path constraints enter as algebraic inequalities [5, 6]. The cost penalises distance to the landing site at every node together with control effort. The problem is built with CasADi [14] and solved with IPOPT [13]. The path constraints are listed in Table 1.

Table 1. Path constraints of the nominal landing problem.

Constraint	Value	Purpose
Glide cone	12° above the horizon	excludes a dive to the surface followed by a crawl to the pad
Attitude	45° per axis	planner envelope
Body rate	10 °/s	planner envelope
Speed	60 m/s per axis and in norm	a per-axis box alone would admit 104 m/s on a shallow approach
Altitude floor	$z > 0$, contact at 1 m	the vehicle may not pass through the surface
Actuator bounds	thrust and gimbal limits	hardware

Two features of the formulation affect the fault results and are stated explicitly. First, a decaying corridor is applied to the post-fault transient. The nominal descent rides its velocity limit exactly, so any perturbation places the post-fault state marginally outside the box, and a hard bound at the first node would report infeasibility for an overshoot of a few millimetres per second. The bounds therefore begin at the larger of the initial excursion and a fixed multiple of the nominal envelope, and contract linearly to the design envelope within a few seconds. Second, the powered problem terminates at a 1 m contact altitude where the engine is cut, and the final stretch is an analytic engine-off ballistic settle. This reproduces the shutdown of the descent engine when the contact probes below the landing gear touch the surface, and keeps the problem well posed near the ground.

C. Success criterion

A trajectory is scored against a fixed landing gate evaluated at touchdown, after the ballistic settle: vertical speed ≤ 3.0 m/s, horizontal speed ≤ 1.2 m/s, tilt $\le 6^\circ$, position error ≤ 15 m and body rate ≤ 5 °/s. The gate margin is the worst of the five criteria normalised by its limit, so a margin ≤ 1 is a landing and the value is a continuous severity measure rather than a pass/fail bit.

A separate loss-of-control test marks states from which no vehicle returns — tilt past 90°, body rates past 120 °/s, gross speed excursions. It is deliberately far outside the planner envelope of Table 1, so that in every case short of it the solver, rather than a chosen threshold, decides whether a recovery exists.

D. Reference descent

All injection campaigns depart from one nominal descent, anchored on the documented Apollo approach-phase (P64) geometry [24, 25]. Two points of that phase are published: high gate at about 7,500 ft, 4.5 nmi and 500 ft/s, and low gate at about 500 ft, 2,000 ft and 60 ft/s, with the approach between them flown essentially straight at the landing point. Interpolating to the fastest state this vehicle model admits with margin gives the anchor state of Table 2.

Table 2. Anchor state of the reference descent, reconstructed from the published Apollo approach geometry [24, 25].

Quantity	SI	Imperial
Altitude	736 m	2416 ft
Range to pad	2724 m	8937 ft
Horizontal speed	55.0 m/s	180 ft/s
Descent rate	15.6 m/s	51 ft/s
Line-of-sight depression	15.13°	
Flight-path angle	15.87°	

The last two rows are a consistency check: a vehicle flying the approach points its velocity vector at the landing site, and the two angles agree to 0.75°. The anchor is a reconstruction of published geometry and not telemetry from any particular mission. The healthy vehicle reaches contact at $t = 65$ s with a gate margin of 0.60. The real P64 approach decelerates continuously, whereas this trajectory holds the speed cap and brakes late; that is a consequence of the cost function, not a property of Apollo. The profile and the injection points used later are shown in Figure 1.

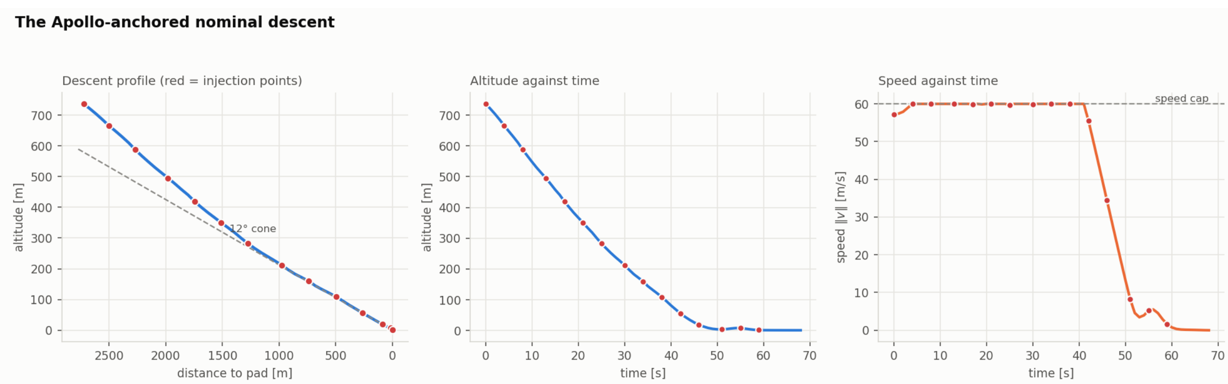

Figure 1. Reference descent, with the injection points marked.

VI. Evaluation campaigns

Four campaigns of optimal-control solves are run on the Lunar Module model, each holding a different variable fixed. They are summarised in Table 3.

Table 3. Design of the evaluation campaigns.

Campaign	Variable studied	Design	Solves
G	Fault class against initial condition	16 faults × 5 regimes × 10	800
F	Engine half-spacing	12 plants × 48 states	576
D, E	Reaction delay and fault severity	bisection on τ_d and η; Sobol sample	272 + 220
H	Injection point along the reference descent	12 faults × 15 points	180
H-R	Constraint relaxation of the H failures	12 cases × 4 corridors × seeds	≤ 22 per case

Campaign G samples initial conditions from five regimes — an approach box, a dispersed box, a low-and-late box, an upset box and a deliberately critical box — and applies each fault of the catalogue to each. Campaign F holds the fault set fixed and varies the engine half-spacing across identical initial conditions, which isolates the effect of actuator geometry. Campaigns D and E inject a fault at time t_f and withhold the planner's response for a delay τ_d, during which the vehicle flies a stale command on a damaged engine; bisection on τ_d and on the severity η locates the critical values, and a Sobol-sampled companion measures the drop in landing rate directly. Campaign H injects each fault at points along the single reference descent, so that every outcome carries a time, an altitude, a range and a speed. Campaign H-R re-solves the failures of campaign H under progressively weaker path constraints, stopping at the first corridor that admits a trajectory; the altitude floor, the actuator bounds and the landing gate are never relaxed, so the weakest corridor leaves $z > 0$ as the only constraint on the state.

Initial-condition boxes are sampled with Sobol sequences [28]. Proportions are reported with Wilson score intervals [29]. Paired comparisons between plants flown from identical initial conditions use McNemar's test [30].

VII. Results

A. Actuation and redundancy across the fleet

Figure 2 shows the per-axis linear and angular acceleration available to each vehicle, grouped by mission class. Three architectural patterns appear, and they cut across the classes. Boosters carry one or a few large gimballed engines and no vehicle-level reaction-control system; their moment authority is the gimbal deflection times the moment arm, and a single centreline engine provides no roll authority at all. Several satellites and Voyager 1 carry only a thruster tier, so every force they produce is coupled to a moment. Crewed vehicles and large probes carry both tiers.

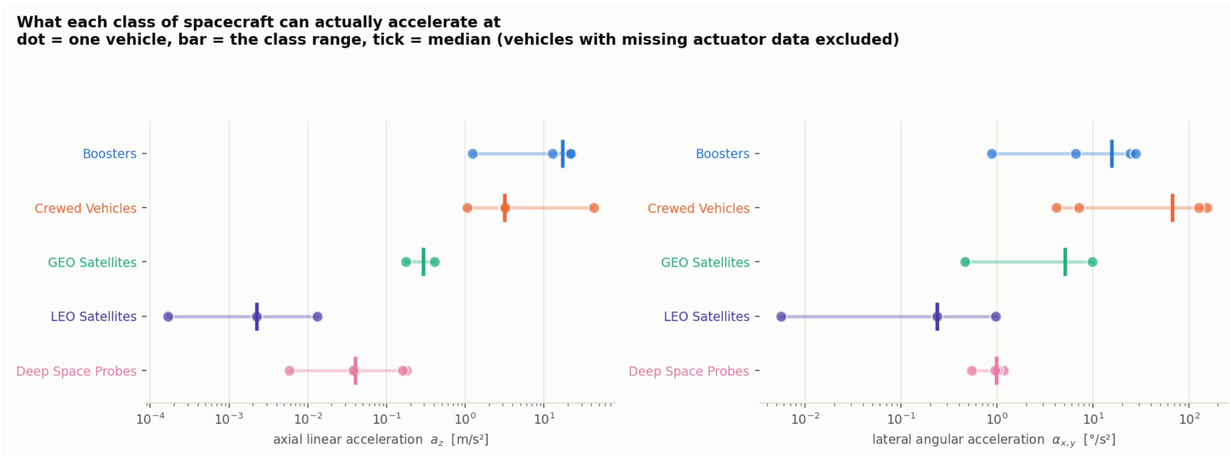

Figure 2. Per-axis linear and angular acceleration by spacecraft class.

Applying the Farkas test to the Lunar Module's 16 thrusters, which are responsible for the three moment axes because a gimballed engine performs the translation, four of the 120 possible losses of two thrusters cost a control degree of freedom; the remainder are absorbed. Enumerating all 2^N loss sets for every vehicle gives the geometry-only failure share plotted in Figure 3. Orion, with 24 thrusters and no fatal loss set smaller than 4 units, retains 70% of its loss space; the Lunar Module, whose smallest fatal loss sets are pairs, retains 28%. Vehicles with a single engine sit at exactly 0.5 on the engine tier, because a single unit admits only two loss sets.

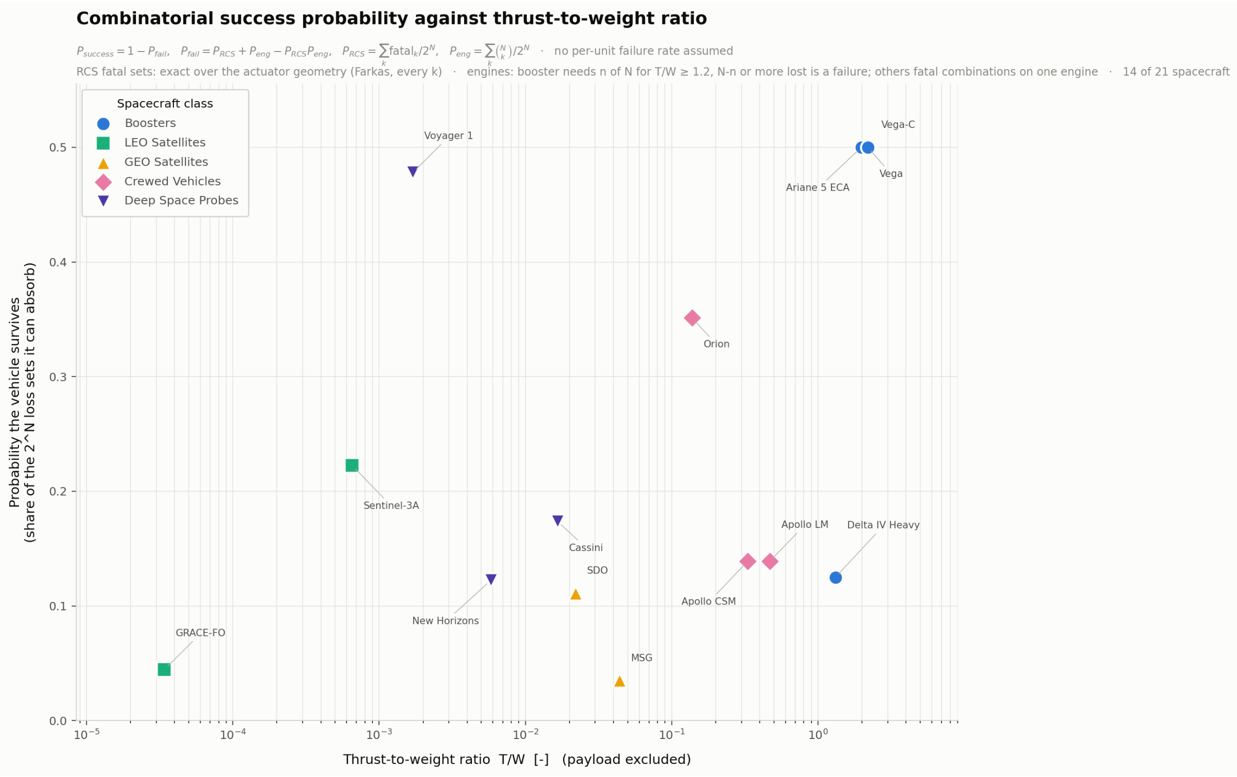

Figure 3. Combinatorial success share against thrust-to-weight ratio.

B. Capability of the Apollo Lunar Module under fault

Each fault of Appendix B was applied to the actuator model and the attainable sets recomputed. Figure 4 shows the resulting measures and Figure 5 the sets themselves. The classification in Table 4 follows from those numbers.

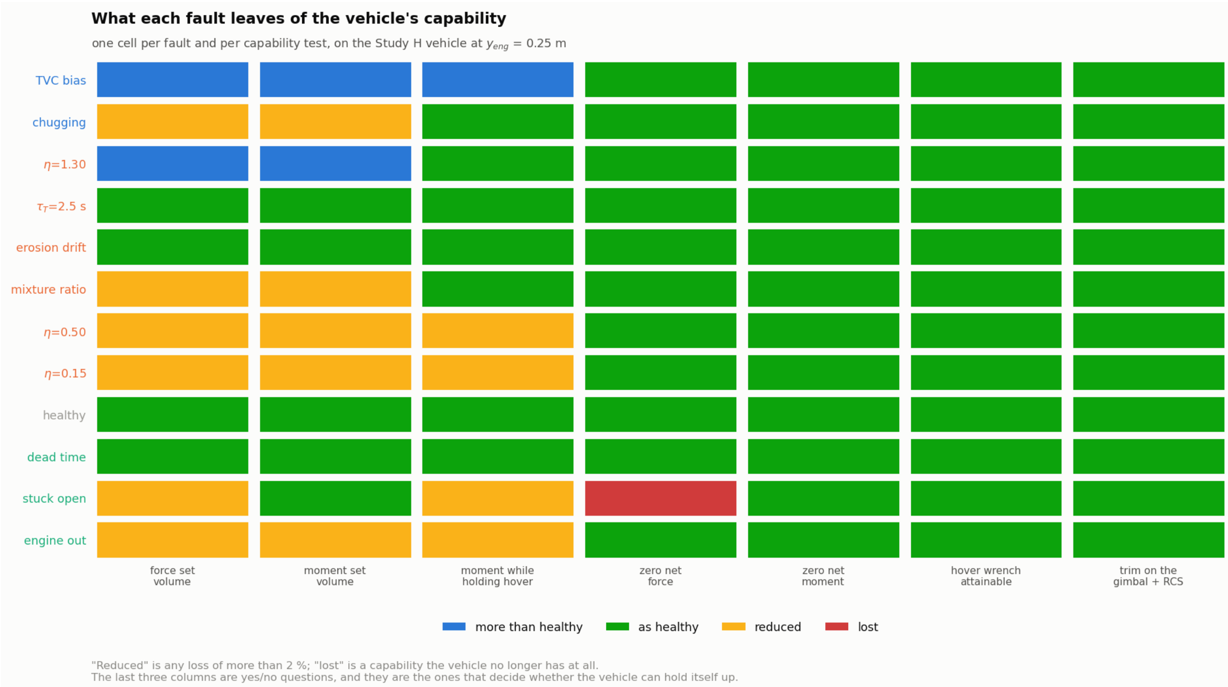

Figure 4. Capability measures retained under each fault, Apollo Lunar Module.

Table 4. Classification of the faults by their effect on the attainable sets.

Effect on the sets	Faults	Definition
Contraction	chugging, η=0.50, η=0.15, mixture ratio, engine out	the damaged set is contained in the healthy one
Expansion	TVC bias, η=1.30	the set grows, but asymmetrically
Loss of the origin	stuck open	the zero wrench becomes unattainable
No change	τ_T=2.5 s, erosion drift, dead time	the sets are numerically identical to the healthy vehicle

The loss-of-origin case is distinct from the others and is invisible to any size measure. For every other fault the engine can be shut down, so the zero wrench is available and the hull contains the origin. A valve stuck open removes that option: the engine's contribution becomes the shell $\{T\mathbf{d} : T \in [T_{min}, T_{max}]\}$ with $T_{min} > 0$, whose hull excludes the origin, while retaining 97% of the healthy force-set volume.

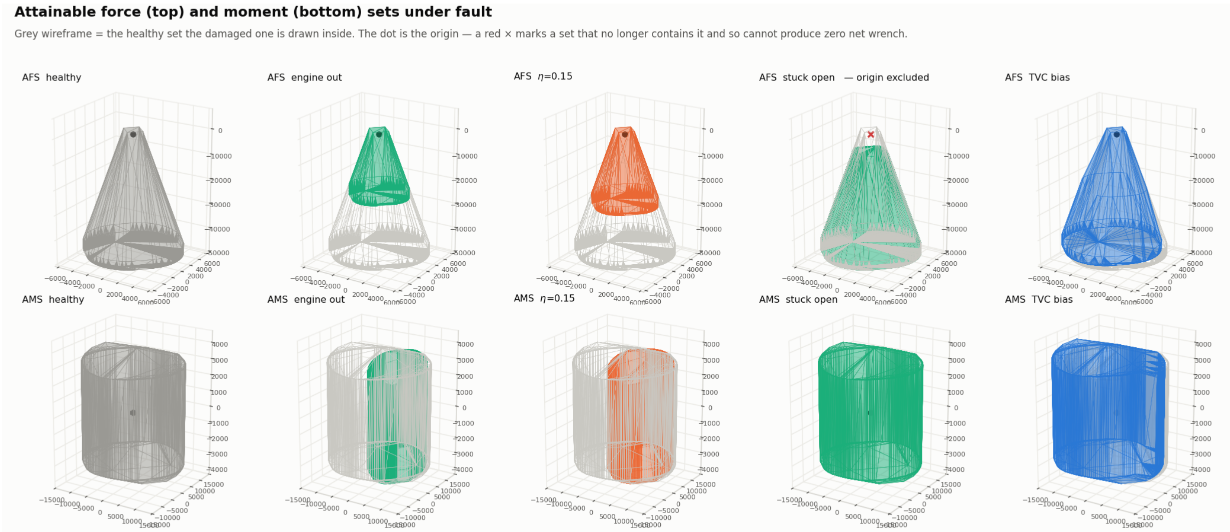

Figure 5. Attainable force sets (top) and moment sets (bottom) under fault.

Table 5 reports the retained fraction of the force set, the moment set and the conditional moment set at the hover thrust of 12,530 N, together with whether the hover wrench itself remains attainable. The conditional measure separates faults that the unconditional sets report as nearly identical. The hover wrench is attainable under every fault examined.

Table 5. Retained fraction of each attainable set under fault, Apollo Lunar Module. Values are relative to the healthy vehicle.

Fault	Effect	AFS	AMS	AMS | hover	Hover wrench
engine out	contracted	0.23	0.28	0.53	yes
η=0.15	contracted	0.30	0.36	0.65	yes
stuck open	origin lost	0.97	1.00	0.68	yes
η=0.50	contracted	0.52	0.59	0.94	yes
mixture ratio	contracted	0.74	0.78	1.00	yes
healthy	unchanged	1.00	1.00	1.00	yes
chugging	contracted	0.88	0.90	1.00	yes
η=1.30	expanded	1.39	1.30	1.00	yes
τ_T=2.5 s	unchanged	1.00	1.00	1.00	yes
erosion drift	unchanged	1.00	1.00	1.00	yes
dead time	unchanged	1.00	1.00	1.00	yes
TVC bias	expanded	1.08	1.04	1.08	yes

Redundancy within each tier was measured by removing units one at a time. Losing any single one of the 16 thrusters leaves between 88% and 95% of the conditional moment authority. Losing one of the two engines leaves 52%.

Sweeping the engine half-spacing and asking, by linear program, whether zero net moment is attainable while holding hover thrust gives a last trimmable spacing of 0.258 m, against the analytic bound of 0.263 m derived in Section III (Figure 6). With the thruster tier available, engine-out trim is achieved at every spacing tested.

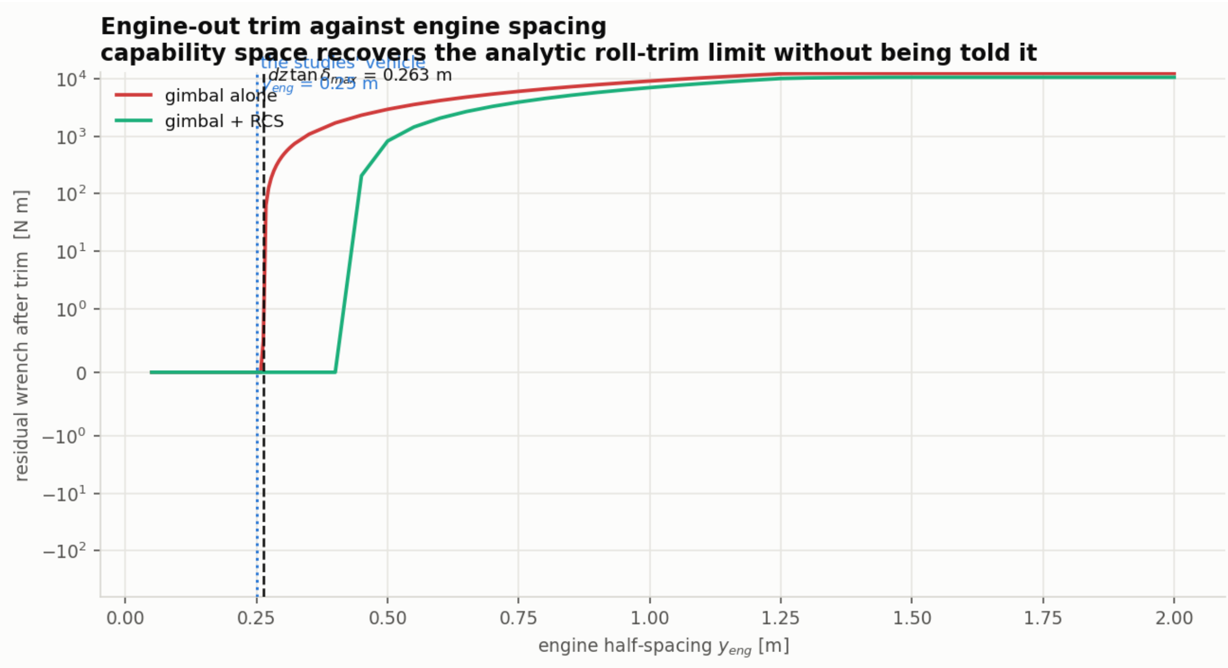

Figure 6. Engine-out trim capability against engine half-spacing.

C. Capability across the fleet

The same measures were applied to all 18 vehicles, using the mean support radius because several moment sets are planar and have zero volume. Figure 7 maps the moment authority retained after the worst single failure of each type, and Table 6 aggregates it.

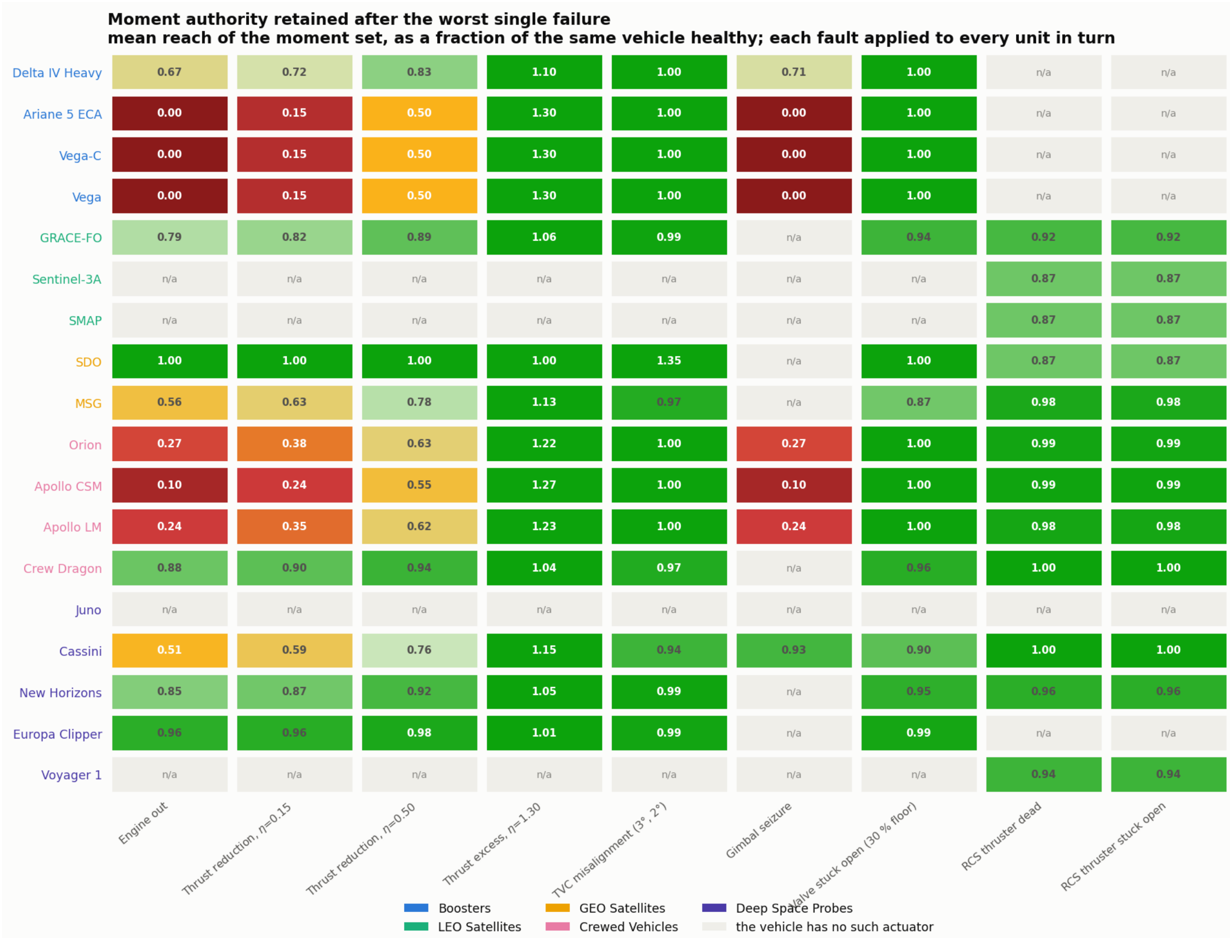

Figure 7. Moment authority retained after the worst single failure, by vehicle and fault.

Table 6. Fraction of moment authority retained across the fleet, worst single failure of each type.

Fault	Vehicles exposed	Mean retained	Range
Engine out	14	0.49	0.00 – 1.00
Thrust reduction, η=0.15	14	0.56	0.15 – 1.00
Thrust reduction, η=0.50	14	0.74	0.50 – 1.00
Thrust excess, η=1.30	14	1.15	1.00 – 1.30
TVC misalignment (3°, 2°)	14	1.01	0.94 – 1.35
Gimbal seizure	8	0.28	0.00 – 0.93
Valve stuck open (30 % floor)	14	0.97	0.87 – 1.00
RCS thruster dead	12	0.95	0.87 – 1.00
RCS thruster stuck open	12	0.95	0.87 – 1.00

The four effects of Table 4 describe every vehicle in the fleet. The magnitudes do not transfer: engine-out spans the full range from total loss of the moment tier to no measurable cost, depending on whether the vehicle's engines are offset from the centreline.

Defining the distance between two vehicles as the mean absolute difference in retained authority over the faults both can suffer, same-class pairs differ by 0.130 and different-class pairs by 0.187 (21 and 100 pairs, Figure 8). The classes are not equally coherent: the low-Earth-orbit satellites respond almost identically to one another, whereas the two geostationary satellites differ by more than two vehicles drawn at random. SDO and MSG are both geostationary and lie at opposite ends of the map; SDO's single engine is on the centreline and MSG's two are not.

Figure 8. Pairwise distance between fault responses, with class membership.

D. Fault class and initial-condition regime

Campaign G comprises 800 solves over 16 faults and 5 regimes. Figure 9 shows the outcome of every cell and Table 7 the landing rate by fault class.

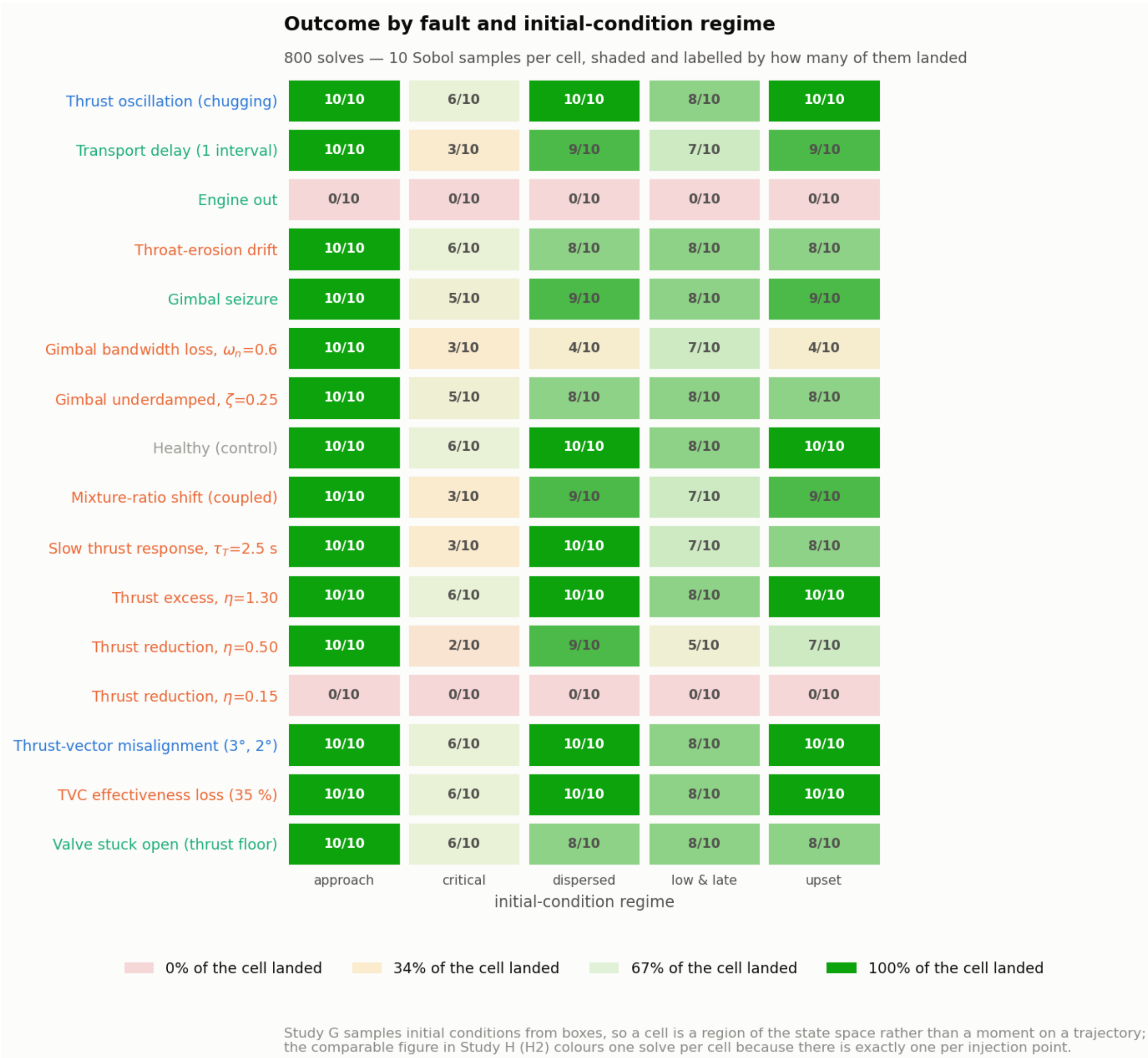

Figure 9. Outcome by fault and initial-condition regime, campaign G.

Table 7. Landing rate by fault class, campaign G.

Fault class	Landed	Rate	95% Wilson interval
none	44/50	0.880	[0.762, 0.944]
additive	88/100	0.880	[0.802, 0.930]
multiplicative	304/450	0.676	[0.631, 0.717]
structural	119/200	0.595	[0.526, 0.661]

Structural faults land at 0.595, multiplicative faults at 0.676 and additive faults at 0.880; the healthy control lands at 0.880. The additive and healthy intervals overlap. Across regimes, the landing rate runs from 0.875 in the approach box to 0.412 in the critical box.

E. Engine half-spacing

Campaign F flies 12 plants from 48 identical initial conditions each, 576 solves in total (Figure 10, Table 8).

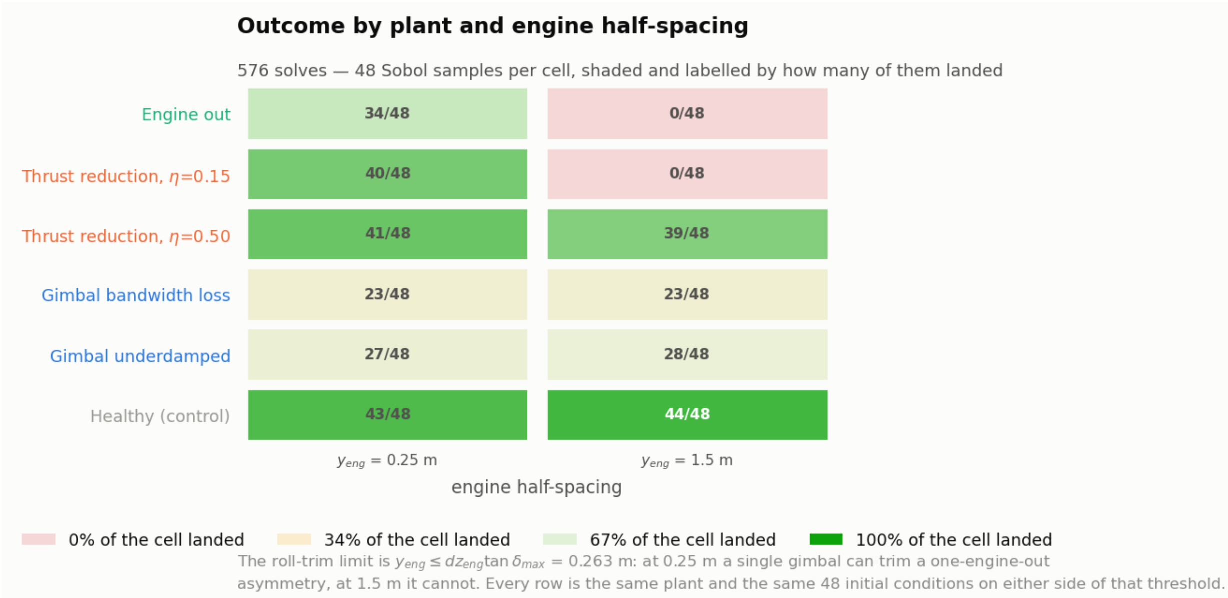

Figure 10. Outcome by plant and engine half-spacing, campaign F.

Table 8. Landing rate at two engine half-spacings, campaign F.

Plant	y_{eng} = 1.5 m	y_{eng} = 0.25 m	Change
healthy	0.917	0.896	-0.021
engine 2 out	0.000	0.708	+0.708
η=0.50	0.812	0.854	+0.042
η=0.15	0.000	0.833	+0.833
gimbal ω_n=0.6	0.479	0.479	+0.000
gimbal ζ=0.25	0.583	0.562	-0.021

The response splits by fault type. Engine-out moves from 0% to 71%, and near-total thrust loss from 0% to 83%. The two gimbal-dynamics faults move by one percentage point or less.

F. Reaction delay and fault severity

Of the 272 solves in campaign D, 57 land, 8 miss the gate, 172 admit no feasible re-plan at the end of the delay, and 35 are lost during the delay itself.

The companion campaign E over 220 Sobol-sampled states gives a landing rate of 0.977 for the healthy vehicle [0.948, 0.990] against 0.205 for the faulted vehicle [0.156, 0.264]. A decision-tree surrogate fitted to those outcomes reaches an AUC of 0.822, with permutation importance concentrated on the fault severity η (0.179) and the onset time t_f (0.040); no component of the vehicle state exceeds either.

G. Injection point along the reference descent

Campaign H injects 12 faults at 15 points along the reference descent, 180 solves, of which 168 land (Figure 11). The faults fall into three groups: those that survive every injection point, those with a boundary part-way along the descent, and those that fail at most of the injection points. Four faults have at least one injection point with no recovery. Figure 12 shows the landing share by fault class against injection time: the failures are concentrated late in the descent, in the braking phase, where the vehicle is low and still fast.

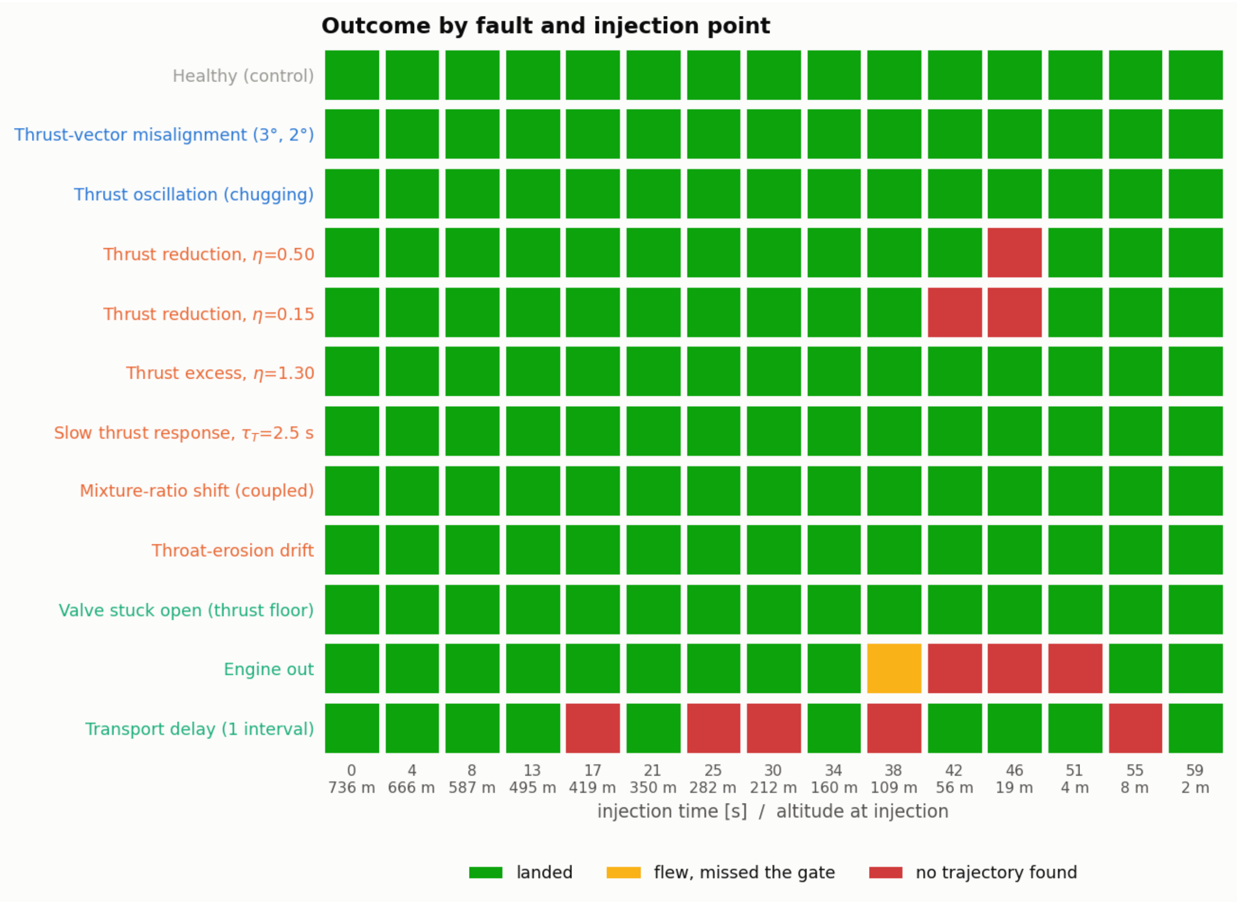

Figure 11. Outcome by fault and injection point along the descent, campaign H.

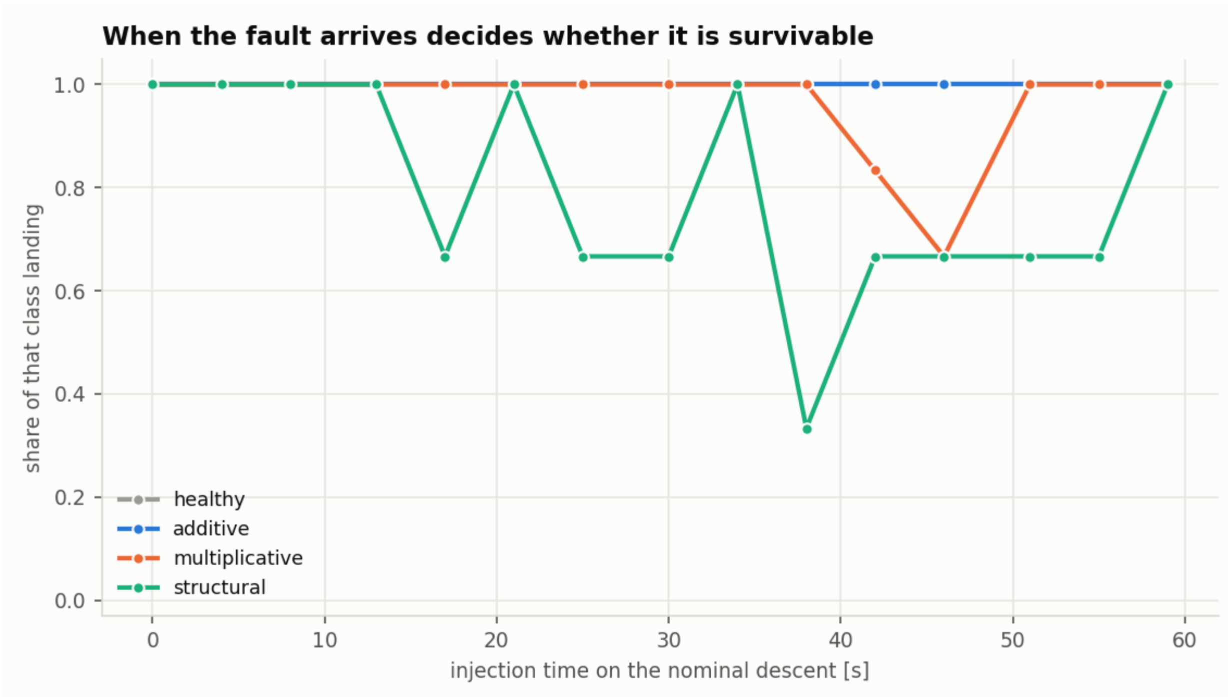

Figure 12. Landing share by fault class against injection time.

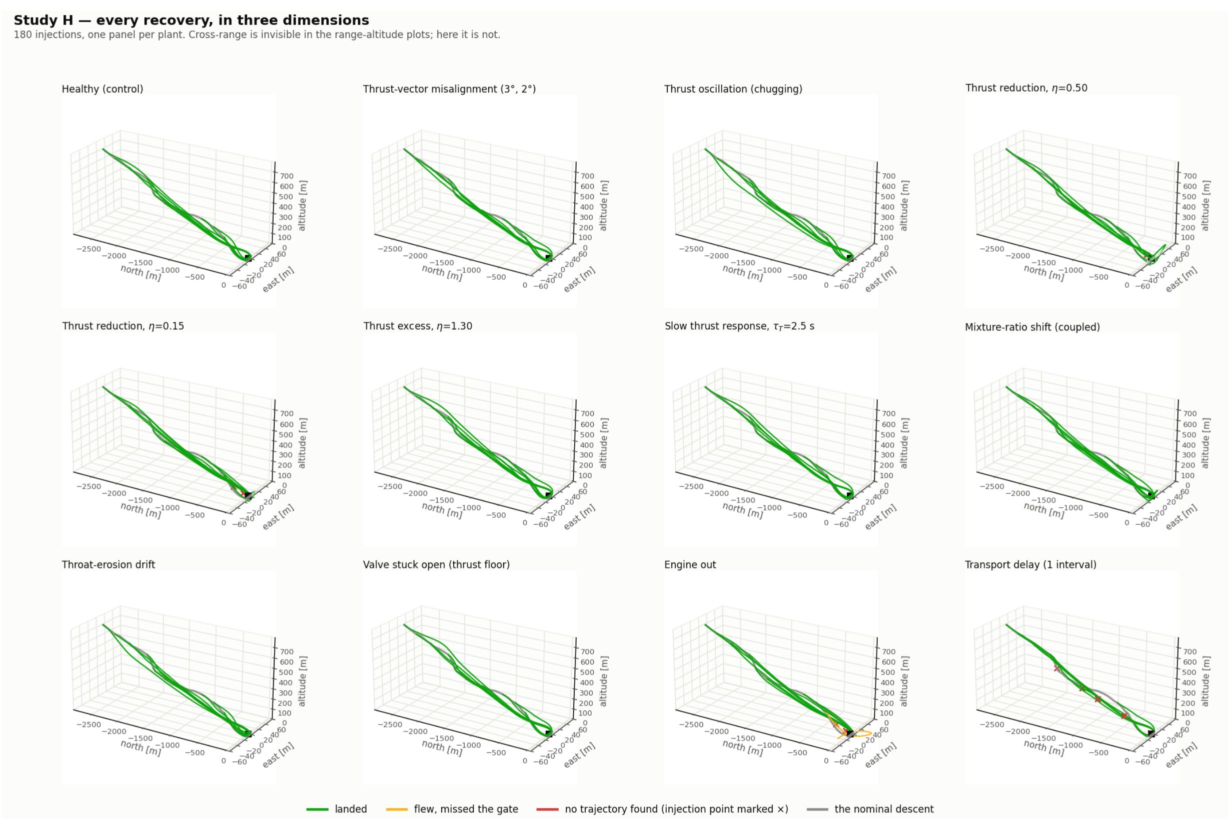

Figure 13. Recovery trajectories in three dimensions, one panel per fault.

Figure 13 shows the recoveries in three dimensions. The asymmetric faults do not remain in the vertical plane: an engine-out vehicle yaws under asymmetric thrust and recovers through a cross-range excursion that returns to the landing site.

H. Constraint relaxation

Campaign H recorded 11 injections with no recovery trajectory. Each was re-solved at four progressively weaker corridors (Figure 14), the weakest of which enforces only $z > 0$ on the state.

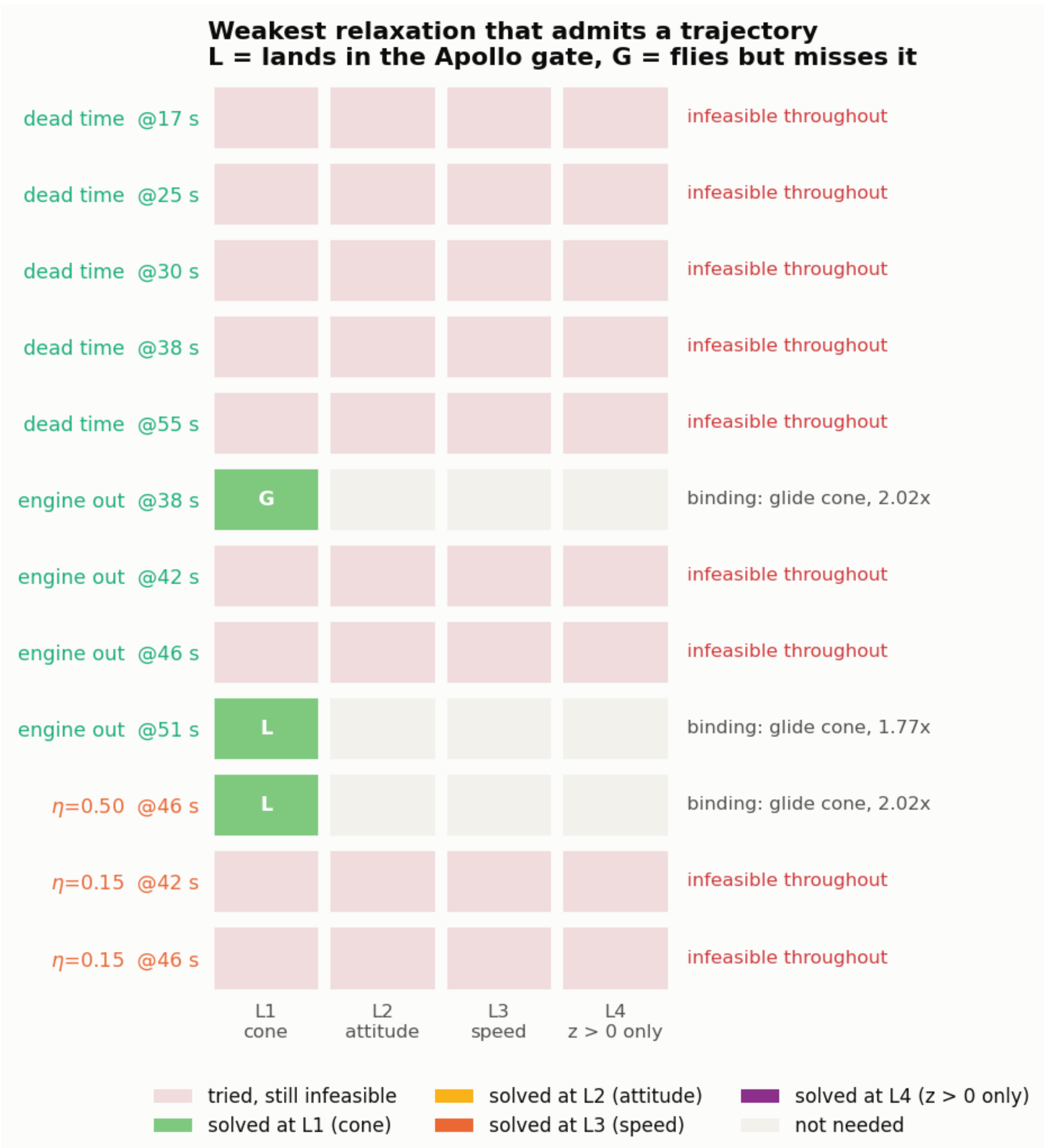

Figure 14. Weakest constraint relaxation that admits a trajectory.

Three of the twelve cases recover, all at the first rung, in which only the glide cone is opened. The attitude, rate and speed relaxations admit no additional case. The recoveries that need the cone fly a 6.0–6.8° path against the 12°

nominal floor.

Nine cases remain infeasible with only $z > 0$ enforced, after 22 independent solver seeds across the four corridors. By fault, these are dead time (5), engine out (2), η=0.15 (2).

VIII. Discussion

A. Attainable sets do not predict survivability

Section VII reports two facts that stand in tension. Every fault examined leaves the Lunar Module's hover wrench attainable (Table 5), and four of them have onset points from which no landing trajectory exists (Figure 11). Figure 15 plots the two measures against each other.

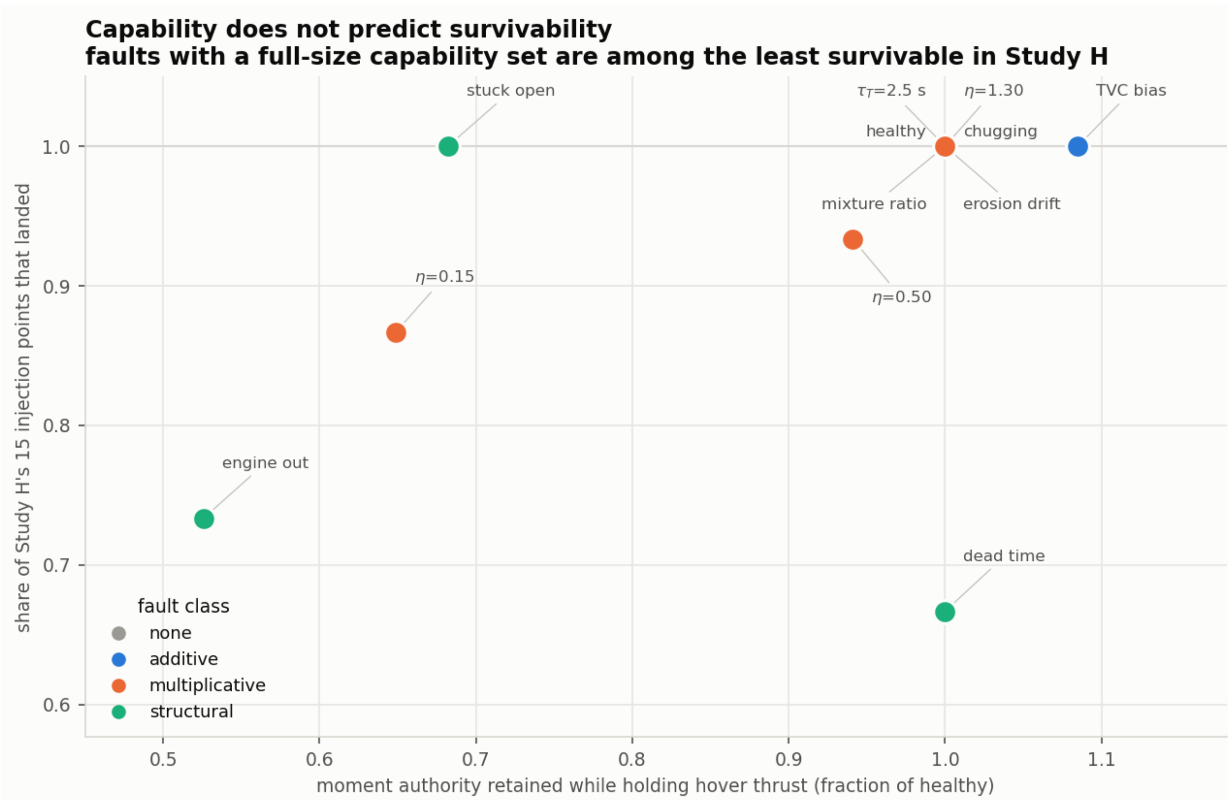

Figure 15. Retained conditional moment authority against landing share.

The transport delay is the clearest case. Its attainable sets are numerically identical to the healthy vehicle in every measure computed — volume, per-axis authority, containment of the origin and attainability of the hover wrench — and it is the least survivable fault in the campaign, landing from 67% of injection points and accounting for five of the nine injections that no relaxation recovered. A delay does not remove authority; it removes the ability to apply that authority at the moment the state requires it. An attainable set is a statement about an instant and cannot represent this.

The converse also holds. The stuck-open valve retains only 68% of its conditional moment authority and loses the origin outright, yet lands from 100% of injection points. A reduced set is survivable as long as what remains contains everything the trajectory demands.

The two measures are therefore complementary rather than redundant. A fault that places the required wrench outside the attainable set is unrecoverable for a reason no control law can address, and allocation analysis identifies it cheaply. A fault that leaves the required wrench inside the set may still be unrecoverable through timing or through the state the vehicle happens to occupy, and only a trajectory campaign identifies it. A fault-tolerant scheme that reasons solely about attainable sets will rank the temporal faults as harmless, because in capability space they are.

B. Two mechanisms of redundancy

A thruster ring and an engine cluster are different redundancy problems, and the distinction holds at both scales examined. On the Lunar Module, losing one of 16 thrusters costs at most 12% of the conditional moment authority, whereas losing one of two engines costs 48%. Across the fleet, thruster-tier faults are the mildest column of Figure 7 for every vehicle and engine-tier faults the harshest. A ring of many small one-sided units degrades gradually; a cluster of few large units degrades in steps.

This also accounts for the fleet similarity result. The separation between same-class and different-class pairs is real but small (0.130 against 0.187), and the vehicles that respond alike share a tier structure rather than a mission. Two geostationary satellites lie at opposite ends of the map because one has its engine on the centreline and the other does not, while a booster and a probe of the same single-engine architecture lie together. For predicting fault response, actuator architecture is the useful classification and mission class is not.

C. Geometry buys tolerance to geometric faults only

Campaign F isolates the effect of actuator geometry. Moving the engines inside the roll-trim threshold of Section III takes engine-out from 0% to 71% landings and leaves the gimbal-dynamics faults where they were. A change to actuator geometry buys tolerance to faults that are themselves geometric. Faults that are temporal — delay, lag, oscillation, drift — are untouched by it. They are also the faults that the attainable sets cannot distinguish from healthy, and the faults that the constraint relaxation of campaign H-R could not recover. The three results are consistent with a single division of the fault space into geometric and temporal effects.

D. Constraints belong in the statement of a result

Campaign H-R reduced 11 reported failures to nine by giving the same states and the same plants a wider corridor, and the only constraint that mattered was the glide cone. Any study reporting fault survivability from a constrained planner is reporting its constraints as much as its vehicle. The defensible form of such a claim is not that a fault is unrecoverable after a given time, but that it is unrecoverable after that time within a stated corridor, and after some later time with the corridor opened. The interval between the two is the value available to an envelope-expanding guidance mode.

E. Limitations

Five assumptions bound the results.

Detection is instantaneous and perfect everywhere except campaigns D and E. Every survivability figure is therefore an upper bound on what a system with a diagnosis latency achieves, and campaign D indicates the penalty is not small.

The planning is open loop. Each result is a single optimal-control solve rather than a closed loop re-solving in flight, which treats evolving faults — the erosion drift in particular — more favourably than a receding-horizon implementation would.

Local infeasibility is not global infeasibility. The transcribed problem is non-convex, so a converged solve proves that a trajectory exists while a failed solve does not prove that none does. This is why campaign H-R uses 22 seeds per case before recording a loss.

The fleet geometry is largely estimated. Of 175 geometry entries, 82 are estimated from published envelopes rather than sourced, and only the Lunar Module layout is validated. The fleet comparisons are comparisons between models of these spacecraft.

Mass is constant. The plant does not deplete propellant, which understates the cost of a partial-thrust fault over a long burn.

IX. Conclusions

Actuator faults were characterised in two independent ways on the same vehicles and the same fault set: as perturbations of the attainable force and moment sets, and as changes in the existence of a feasible landing trajectory.

Four effects on the attainable sets account for every fault and every vehicle examined: contraction, expansion, loss of the origin, and no change. The classification transfers across 18 spacecraft in five mission classes; the magnitudes do not, and depend on actuator architecture rather than on mission class.

The two characterisations do not agree. Every fault studied leaves the Lunar Module's hover wrench attainable, yet four of them have onset points from which no landing trajectory exists, and the least survivable fault of all leaves the attainable sets numerically unchanged. Attainable-set analysis is necessary but not sufficient for fault-tolerant design.

Two mechanisms of redundancy are distinguished by the same data. A ring of many one-sided thrusters degrades gradually, at most 12% of conditional moment authority per unit lost, whereas a cluster of two engines degrades in steps of 48%. Redundancy is a geometric property, not a count: four of the 120 ways to lose two Lunar Module thrusters cost a degree of freedom, and the rest are absorbed.

Actuator geometry buys tolerance to geometric faults and none to temporal ones, and the temporal faults are precisely those the attainable sets cannot see. The natural extension of this work is therefore to replace the attainable set, a statement about an instant, with its reachable set over a horizon, and to pair it with a detection model and a closed-loop re-planner so that the upper bounds reported here can be converted into achievable performance.

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Appendix A: Origin of each result

Every number and figure in this document was read from the study that produced it; no result was recomputed for this document.

Table 9. Mapping from each result to the script that produced it.

Result	Produced by	Section
Fleet actuation envelopes, per-axis accelerations	`studies/actuation_envelopes/`	VII.A
Thruster-loss enumeration, combinatorial shares	`studies/reliability/`	VII.A
Engine-placement threshold	`studies/engine_placement/`	III.D
Capability space, single vehicle	`studies/capability_space/run_capability_study.py`	VII.B
Capability space, fleet	`studies/capability_space/run_fleet_capability.py`	VII.C
Vehicle model and landing OCP	`src/apollo_gnc/apollo_full.py`	V
Fault library, gate and loss criteria	`studies/fault_onset/fault_lib.py`	IV, V
Campaign G	`studies/fault_taxonomy/`	VII.D
Campaign F	`studies/fault_onset/run_study_F.py`	VII.E
Campaigns D and E	`studies/fault_onset/run_study_D.py`, `run_study_E.py`	VII.F
Campaign H	`studies/fault_injection/run_injection_study.py`	VII.G
Campaign H-R	`studies/fault_injection/run_relaxed_study.py`	VII.H

Appendix B: Fault catalogue

The faults evaluated on the Apollo Lunar Module, with the class of Section IV, the section of the underlying fault framework each is drawn from, and its temporal character.

Table 10. Fault catalogue evaluated on the Apollo Lunar Module.

Fault	Class	Framework §	Temporal character
Healthy (control)	none	—	—
Thrust-vector misalignment (3°, 2°)	additive	2.7	incipient
Thrust oscillation (chugging)	additive	2.3	intermittent / forced
Thrust reduction, η=0.50	multiplicative	2.1	abrupt / incipient
Thrust reduction, η=0.15	multiplicative	2.1	abrupt
Thrust excess, η=1.30	multiplicative	2.2	abrupt
Slow thrust response, τ_T=2.5 s	multiplicative	2.5	incipient
Mixture-ratio shift (coupled)	multiplicative	2.4	incipient
Throat-erosion drift	multiplicative	2.10	incipient
Valve stuck open (thrust floor)	structural	2.2 / 1.4	abrupt
Engine out	structural	2.1 / 3.1	abrupt
Transport delay (1 interval)	structural	2.6	abrupt / intermittent